Sanjivani Rural Education Society's

Sanjivani College of Engineering, Kopargaon-423603

(An Autonomous Institute Affiliated to Savitribai Phule Pune University, Pune)
NAAC 'A' Grade Accredited

Department of Information Technology

UG Programme in IT NBA Accredited

FINAL YEAR PROJECT SYNOPSIS

(Academic Year 2026-2027)

Project Group Id	
Title of the Project	Smart college FAQ system
Project Domain	Natural Language Processing (NLP) / Information Retrieval
Problem Statement	To develop an NLP-based smart FAQ system that automatically understands students' and staff members' queries and provides relevant answers related to college departments, academics, examinations, admissions, fees, timetable, placements, facilities, and other common college-related information.
References	1. E. Adamopoulou and L. Moussiades, "An Overview of Chatbot Technology," in *Artificial Intelligence Applications and Innovations*, Springer, 2020. 2. A. Følstad and P. B. Brandtzaeg, "Chatbots and the New World of HCI," *Interactions*, vol. 24, no. 4, 2017, pp. 38–42. 3. J. Hill, W. R. Ford, and I. G. Farreras, "Real Conversations with Artificial Intelligence: A Comparison Between Human–Human Online Conversations and Human–Chatbot Conversations," *Computers in Human Behavior*, vol. 49, 2015, pp. 245–250. 4. D. Jurafsky and J. H. Martin, *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition*, 3rd ed., 2024. 5. S. K. Shukla, A. K. Singh, and others, "A Survey on Question Answering Systems Using Natural Language Processing," *International Journal of Computer Applications*, relevant literature on NLP-based question answering systems.
Project Plan	Week 1 - Requirement Identification and Literature Survey Finalize the project title, define the problem statement, objectives, scope, and target users. Identify common college-related queries such as admissions, academics, examinations, fees, timetable, placements, departments, and facilities. Conduct a preliminary literature survey on NLP-based FAQ and question-answering systems and identify suitable technologies. Week 2 - Dataset Collection and Preprocessing Collect or create a college FAQ dataset containing questions and their corresponding answers. Categorize FAQs into different topics such as Admissions, Academics, Examination, Fees, Placements, Hostel, Library, and General Information. Perform data cleaning, duplicate removal, text normalization, tokenization, stopword removal, and other required NLP preprocessing.

	Week 3 - NLP Model and FAQ Retrieval Development Convert FAQ questions into numerical/vector representations using techniques such as TF-IDF, CountVectorizer, or text embeddings. Develop a question-matching or similarity-based retrieval system using Cosine Similarity to identify the most relevant FAQ and return its corresponding answer. Week 4 - System Evaluation and Application Development Evaluate the system using measures such as Accuracy, Precision, Recall, F1-score, and response relevance. Test the system with different types of student queries and improve the matching process. Develop a basic web-based application that allows users to enter questions and displays the most relevant answer.
Name of Students in the Project Group with their signatures	1.Onkar Gunjal 2.Chaitanya Gurav 3.Ritesh Gursal 4.Abhi Hajare
Name of the Guide and Signature	Dr. Sachin Thanekar
Date	

Project title Accepted.

Mr. Sachin Thanekar	Dr. Sachin Thanekar	Dr. M A Jawale
Subject Coordinator	Project Guide	HOD IT & CS